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Whole-genome sequencing of *Klebsiella pneumoniae* MDR circulating in a pediatric hospital setting: a comprehensive genome analysis of isolates from Guayaquil, Ecuador

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Abstract

Background *Klebsiella pneumoniae* is the major cause of nosocomial infections worldwide and is related to a worsening increase in Multidrug-Resistant Bacteria (MDR) and virulence genes that seriously affect immunosuppressed patients, long-stay intensive care patients, elderly individuals, and children. Whole-Genome Sequencing (WGS) has resulted in a useful strategy for characterizing the genomic components of clinically important bacteria, such as *K. pneumoniae,* enabling them to monitor genetic changes and understand transmission, highlighting the risk of dissemination of resistance and virulence associated genes in hospitals. In this study, we report on WGS 14 clinical isolates of *K. pneumoniae* from a pediatric hospital biobank of Guayaquil, Ecuador.

Results The main findings revealed pronounced genetic heterogeneity among the isolates. Multilocus sequencing type ST45 was the predominant lineage among non-KPC isolates, whereas ST629 was found more frequently among KPC isolates. Phylogenetic analysis suggested local transmission dynamics. Comparative genomic analysis revealed a core set of 3511 conserved genes and an open pangenome in neonatal isolates. The diversity of MLSTs and capsular types, and the high genetic diversity among these isolates indicate high intraspecific variability. In terms of virulence factors, we identified genes associated with adherence, biofilm formation, immune evasion, secretion systems, multidrug efflux pump transporters, and a notably high number of genes related to iron uptake. A large number of these genes were detected in the ST45 isolate, whereas iron uptake yersiniabactin genes were found exclusively in the non-KPC isolates. We observed high resistance to commonly used antibiotics and determined that these isolates exhibited multidrug resistance including β-lactams, aminoglycosides, fluoroquinolones, quinolones, trimetropins, fosfomycin and macrolides; additionally, resistance-associated point mutations and cross-resistance genes were identified in all the isolates. We also report the first *K. pneumoniae* KPC-3 gene producers in Ecuador.

Conclusions Our WGS results for clinical isolates highlight the importance of MDR in neonatal *K. pneumoniae* infections and their genetic diversity. WGS will be an imperative strategy for the surveillance of *K. pneumoniae* in Ecuador, and will contribute to identifying effective treatment strategies for *K. pneumoniae* infections in critical units in patients at stratified risk.

Keywords *K. pneumoniae*, Resistance, Virulence, Ecuador, Whole-genome sequencing

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Background

Klebsiella pneumoniae, a rod-shaped bacterium from the Enterobacteriaceae family, is a Gram-negative, nonmotile, and encapsulated microorganism. Historically, this bacterium has been a significant cause of nosocomial infections, predominantly affecting immunosuppressed patients, particularly in UCI settings, elderly individuals, patients with diabetes mellitus, alcoholics, and pediatric patients [1–3]. In 2019, the Global Burden of Diseases, Injuries, and Risk Factor Region (GBD), based on deaths attributable to antimicrobial resistance (AMR), reported 1.27 million deaths worldwide; from these deaths, the mortality caused by *K. pneumoniae* infections contributed to nearly 20% of the deaths [4]. Antibiotic resistance in *K. pneumoniae* has evolved through diverse mechanisms including plasmid acquisition, biofilm formation, cross-resistance patterns, and resistance-associated point mutations [5]. *K. pneumoniae* is a growing cause of mortality, and its classification by the World Health Organization as a priority drug-resistant organism [6] underscores the urgent need to better understand its virulence mechanisms.

The virulence of *K. pneumoniae* is attributed to a plethora of factors, including glycan antigens such as capsule polysaccharides (CPS) and lipopolysaccharides (LPSs), adhesins, the iron uptake system, secretion systems, and biofilm formation, which allow it to evade the host's innate immune mechanisms [7]. These factors can be compounded by the concomitant emergence of multidrug-resistant (MDR) isolates [8]. Owing to their high virulence, antimicrobial resistance, and ability to evade the immune system, these bacteria have been recognized as part of the "ESKAPE" group [9–11], which has become a serious public health concern because of their high morbidity and mortality rates, mainly neonatal infections. Furthermore, infections caused by *K. pneumoniae* carbapenemase (KPC)-producing bacteria (KPC-KP) and *K. pneumoniae* producing extended-spectrum-β-lactamase (ESBL-KP) are especially dangerous, mainly because of the Bradley mechanism of multiresistance and a decrease in antibody efficacy [12–16]. Understanding the virulence mechanisms of *K. pneumoniae,* given its high mortality rate and resistance, represents a significant challenge in the management of infections, especially in vulnerable environments such as pediatric patients in Ecuador.

In Ecuador, the emergence of KPC-producing *K. pneumoniae* has posed a significant threat to healthcare settings since the first report of a *K. pneumoniae* harboring the *bla*$_{\mathrm{KPC\text{-}2}}$ gene in 2013 [17]. Subsequent investigations based on conventional genetic testing, including multiplex PCR and MLST, have demonstrated the rapid spread of KPC-KP and ESBL-KP throughout the country, particularly because of the high prevalence of *K. pneumoniae bla*$_{\mathrm{KPC}}$, ST25, and ST258 [18–20]. Notably, during the period from to 2018–2021, the National Surveillance Antimicrobial Resistance Program in Ecuador identified *K. pneumoniae* as the main pathogen associated with healthcare-associated infections (HAIs), and KPC-KP was determined to be widely distributed across the country [21].

Although traditional phenotypic and genotypic approaches have provided valuable insights, Whole-Genome Sequencing (WGS) has emerged as a powerful molecular diagnostic tool that is widely employed in research. WGS can predict drug resistance, trace lineage, and transmission and define outbreaks of *K. pneumoniae* [12, 22, 23]. In 2020, WGS revealed a carbapenem-resistant isolate harboring *bla*$_{\mathrm{OXA\text{-}48}}$-like gene in Ecuador [24]. The rapid spread of antimicrobial resistance in *K. pneumoniae* is linked to the variability of virulence genes and swift dissemination within intrahospital environments, resulting in KPC-KP being more prevalent than the other resistance types. Despite efforts to monitor this pathogen in the country, few studies have genetically characterized outbreaks using WGS. In this study, we aimed to characterize *K. pneumoniae* isolates from a pediatric hospital in Guayaquil using comparative genome analyses and determine their genome composition.

Methods

Clinical isolates

From a total of 28 Extended-spectrum β-lactamases (ESBLs) of *K. pneumonia* from the biobank collected between 2020 and 2021 in the pediatric Hospital Roberto Gilbert in Guayaquil, 14 isolates were randomly selected for this study. The expanded clinical data of the isolates are shown in Table 1. Antimicrobial susceptibility testing was performed with ampicillin-sulbactam (SAM), ceftriaxone (CRO), gentamicin (GEN), cefepime (FEP), ciprofloxacin (CIP), doripenem (DOR), imipenem (IPM), meropenem (MEM), piperacillin-tazobactam (TZP), amikacin (AMK), fosfomycin (FOF), cefotaxime (CTX), nitrofurantoin (NIT), and trimethoprim-sulfamethoxazole (SXT) using the VITEK-2 Gram Negative Susceptibility Test Cards (AST-GN04). The classification of resistance and susceptibility to each antibiotic was defined according to CLSI guidelines (CLSI M100 ED34:2024) [25]. Antibiotic susceptibility data are shown in Supplementary Table 1.

DNA extraction and whole-genome sequencing

Genomic DNA was extracted using a QIAamp DNA Kit (Qiagen, U.S.A.) according to the manufacturer's instructions. The purity and quantity of DNA were determined using a Qubit 4.0 fluorometer (Invitrogen, Carlsbad, CA, USA). The sequencing library was

Table 1 Clinical data of *K. pneumoniae* isolates

Isolate	ICUs	Age	Sample type	Date sample collection	Antimicrobial resistance phenotype
KPEC1K	Yes	1 month	Blood culture	**06/10/20**	SAM, CRO, GEN
KPEC2K	Yes	2 months	Blood culture	**06/10/20**	SAM, CRO, GEN
KPEC5K	Yes	3 months	Blood culture	**07/10/20**	SAM, CRO, GEN
KPEC6K	Yes	3 months	Blood culture	**07/10/20**	SAM, CRO, FEP[a], CIP, DOR, GEN, IPM, MEM, TZP
KPEC9K	Yes	1-year	Urine culture	**14/10/20**	FEP[a], SAM, CRO, IPM
KPEC10K	Yes	11 years	Urine culture	**14/10/20**	AMK, FEP, SAM, CRO, CTX, GEN, MEM, NIT, SXT, CIP
KPEC11K	Yes	3 months	Blood culture	**11/10/20**	SAM, CRO, GEN
KPEC13K	Yes	8 months	Blood culture	**11/10/20**	SAM, CRO, GEN, CIP
KPEC14K	Yes	8 months	Blood culture	**11/10/20**	SAM, CRO, GEN, CIP, FEP[a]
KPEC15K	Yes	2 months	Peritoneal fluid culture	**14/10/21**	FEP, SAM, CRO, GEN, IPM, TZP, CIP
KPEC28K	Unknown	Unknown	Unknown	Unknown	Unknown
KPEC29K	Yes	1 month	Blood culture	**14/11/20**	AMK, FEP, SAM, CRO, GEN, IPM, TZP
KPEC34K	Yes	1 month	Blood culture	**11/01/21**	FEP, SAM, CRO, GEN, IPM, TZP
KPEC37K	Yes	1 month	Blood culture	**07/01/21**	AMK, FEP, SAM, CRO, GEN, IPM, TZP

[a] *SDD* Susceptible-dose-dependent

constructed from 100 ng of genomic DNA using a tagmentation-based library prep kit (Illumina, Inc., San Diego, CA, USA) according to the manufacturer's instructions. The sequencing library was quantified before sequencing using the Illumina MiniSeq platform with a High Output Reagent Kit, which produced 150 bp paired-end reads.

Quality reads and genome assemblies

To ensure data accuracy, reads with a Q-score ≥ 35 were included in further analyses. FastQC [26] and Trimommatic 0.39 [27] were used with the default parameters for quality analysis and adapter trimming of paired-end raw reads. De novo assembly and error correction were accomplished using SPAdes assembler v.3.15.5 [28] with a K-mer ranging from K21-K71. Whole-genome alignment and reference scaffolding for the improvement of the draft genomes were conducted by RagTag v.2.1.0 [29] using the reference *K. pneumoniae* HS11286 (accession number NC_016845.1). Contig sequences shorter than 200 bp were eliminated in the draft genomes by BBmap v.38.18 [30]. The quality of the assemblies was assessed using QUAST v.5.0.5 [31] with default settings.

Genome analysis of genetic features

Multilocus Sequence Typing (MLST), O-Locus, and K-type analyses were performed with Kleborate v.2.3.1 [32]. The BLAST Ring Image Generator (BRIG) [33] was used to plot the chromosome map. The PHASTER web server and IslandViewer 4 were used to detect prophages and genomic islands (GIs) [34, 35]. Furthermore, we analyzed the presence or absence of antibiotic resistance genes and plasmid replicon in the *K. pneumoniae* genome assemblies with ABRicate v.1.0.1 [36] local databases of reference were employed: NCBI AMRFinderPlus [37] and PlasmidFinder database (https://cge.food.dtu.dk/services/PlasmidFinder/, accessed on 02 April 2024). Association of KPC, non-KPC, plasmid and virulence were shown using the Sankey plot created by SankeyMATIC software (https://sankeymatic.com/, accessed on 10 April 2024). For plasmid reconstruction, we used BBmap v.38.18 to extract mapped reads from the raw reads and subsequently assembled them using SPAdes v3.15.5. In order to improve the data processing in the server, a specific database was generated based on the identified resistance genes. These databases were subsequently aligned with the raw reads to ensure a minimum depth exceeding 10X (Supplementary Table 2). The Pileup function in BBmap [30] was used for the read-depth calculation, whereas the CARD database v3.2.7 [38] was used for antibiotic family classification. The resistance-mediating mutation genes were determined using ResFinder v.4.0 [39], and the presence of the genes (*ompk, parC, and gyrA*) was confirmed using BLAST [40], with a cutoff of 75% coverage. To identify virulence factor genes, Ridom SeqSphere software v.8.3.1 [41] was used, employing the complete Virulence Factor Database (VFDB). The isolates were grouped based on resistance and virulence factors using the cluster analysis by similarity profiles (SIMPROF) test.

Functional annotation, pangenome, and phylogenetic analysis

The genomes assembled from 14 K. *pneumoniae* isolates were annotated by Prokka v.1.14.6 [42]. The annotated genomes were subjected to pangenome analysis via Roary v3.13.0 [43]. The pan-genome genes were divided into four partitions: core genes present in 99–100% of the isolates, soft-core genes in 99–95%, shell genes in 95–15%, and cloud genes in 15–0% of the isolates [44]. Pangenome graphics were constructed using the pangenome profile analysis tool (PanGP) [45]. Functional and structural annotations were made using the Rapid Annotation tool in the Subsystem Technology toolkit (RASTtk) in the Pathosystems Resource Integration Center (PATRIC) [46]. For phylogenetic reconstructions, core-genome single nucleotide polymorphisms (cgSNPs) were identified using Snippy v4.6.0 [47] and gubbins [48] for highly accurate reconstructions. Our dataset consisted of 412 K. *pneumoniae* genomes, including 398 randomly selected genomes from the Pasteur Institute Database (Supplementary Table 3), encompassing approximately 20% for each continent as geographical references [49, 50] and 14 genomes of Ecuadorian isolates. *K. pneumoniae* MDR clinical isolated HS11286 (accession number NC_016845.1) was used as a reference genome. The cgSNPs were subsequently used to construct a phylogenetic tree with the maximum likelihood (ML) method through FastTree v.2.1.11 [51].

Results

Clinical antibiotic susceptibility characteristics of *K. pneumoniae* isolates

The clinical profiles of the *K. pneumoniae* isolates, including data sample collection, sample type, age, and antimicrobial resistance phenotype, are shown in Table 1. Complete microbiological information was available for 92.8% (13/14) of the isolates, seven of these isolates were resistant to carbapenems. However, in silico genomic analysis revealed that only four of these isolates contained bla_{KPC}. Antimicrobial susceptibility testing via MICs across 14 antibiotics showed that all isolates were resistant to ampicillin-sulbactam, ceftriaxone (100% each), and gentamicin (92%) (Supplementary Table 1). Moreover, the susceptibility of the three isolates to cefepime was dose dependent. Doripenem, cefotaxime, nitrofurantoin, and sulfamethoxazole were found to be effective against most of the isolates tested.

Molecular characteristics of *K. pneumoniae* clinical isolates

The genome size of the 14 K. *pneumoniae* isolates was 5.52 Mb with a GC content of 57.35% and a median N50 of 5,070,563 bp. Coding sequences were around 5.157, and approximately 68 tRNAs were identified. The sequencing depth ranged from 8 to 79X, with an average of 26.56X (Supplementary Table 4). Based on in silico typing and MLST, 8 different STs were identified among the isolates. The most common ST was ST45 (6/14), followed by ST629, ST307, and ST1774. Four loci variants were identified in STs (ST45-2LV from ST45, ST629-2LV, ST629-3LV from ST629, and ST13-2LV) (Supplementary Table 4). Capsular typing showed that K62 and K107 were identified in 9/14 of the isolates, with frequencies of five and four isolates, respectively, while K50 was identified in two isolates, and the remaining three isolates corresponded to K135, K160, and K3 types (Supplementary Fig. 1). Furthermore, we found that K62 is associated with a higher number of virulence genes.

Four O-loci, O3/O3a, O2afg, O2a, and O12, were identified in Ecuadorian isolates; the O2a locus was the most common (7/14), while the O12 locus was the least common (1/14). Among the isolates with the O2a locus, two had low confidence levels (Supplementary Table 5).

Functional annotation

Functional genome annotation using the COG and subsystem databases showed consistent results. For instance, the COG database predicted an average of 2,802 genes associated with subsystems, with isolate KPEC11K having the highest number of genes (2,954 genes) and isolate KPEC37K having the lowest number of genes (2,639 genes). The genes were mostly classified in the superclass associated with metabolism (41.85%), followed by energy (14.3%), membrane transport (8.9%), stress response (8,26%), and protein processing (8.24%) (Supplementary Fig. 2). The cofactor, vitamins, and prosthetic group (313 genes); amino acid metabolism (288 genes); carbohydrate metabolism (254 genes); energy and precursor metabolite generation (246 genes); membrane transport (250 genes); and stress response, defense, and virulence (232 genes) were the most representative classes (Supplementary Fig. 2, Supplementary Table 6, Supplementary Table 7).

Regarding subsystems, 393 structural complexes were identified, with an average classification for each isolate showing predominance in metabolism (34%), stress response, defense and virulence (12.3%), protein processing (12.1%), energy (10.1%), membrane transport (8.6%), cellular process (6.1%), DNA processing (5.2%) among others (experimental subsystem) (Supplementary Table 7).

Genes that are specific to each isolate are of considerable interest because of their potential impact on the variability observed within these isolates. Through a functional annotation analysis, 29 unique genes were identified. These genes are associated with various biological functions, including metabolism (9 genes), membrane transport (7 genes), energy (3 genes), cell envelope

(2 genes), stress response, defense and virulence (2 genes), cellular processes (1 gene), and protein processing (1 gene), as well as an additional category of miscellaneous genes related to P2-like phages, exclusive to the KPEC11K isolate (4 genes) (Fig. 1).

Comparative genomic analysis

A circular genome map was produced to identify regions in the clinical isolates that differed from the KPC clinical isolate as the reference genome (HS11286, accession code NC_016845.1). Overall, the BRIG analysis highlighted that the regions covered in the isolates exhibited high genomic identity with the reference genome in terms of presence, but this did not encompass the additional genomic elements that may be unique to each isolate. The draft genomes of the clinical isolates are represented in a BRIG plot and plotted according to drug resistance. Concentric blue rings represent non-bla_{KPC} isolates, whereas red rings represent bla_{KPC} isolates (Fig. 2). Overall, BRIG showed > 99% identity in the coverage zones of the isolates. Most of the gaps in the circles represent prophage regions remaining in the reference genome that were absent from clinical isolates. Some of the phages identified were found within the Ecuadorian isolates with variable coverage, suggesting that these phages contribute to the observed genomic variability. The prophage with the highest average coverage was prophage 1 (Phage_Bacill-vB_Bts_BMBtp14) (98%), whereas prophages 2 (Phage_Entero_P4), 3 (Phage_Entero_P4), 4 (Phage_Edward_GF_2), 5 (Phage_Klebsi_ST512_KPC3phi13.2), 6 (Phage_Salmon_SEN34), 7 (Phage_Klebsi_ST147_VIM-1phi7.1), and 8 (Phage_Salmon_ST64B) had coverages of 14.46%, 25.73%, 9.97%, 22.94%, 13.13%, 31.78%, and 38.1%, respectively; prophage 4 had the lowest average coverage (Supplementary Table 8). Nevertheless, it must be taken into consideration that PHASTER identified prophages 1, 3, and 8 as incomplete; prophages 4, 5, 6, and 7 as intact; and prophage 2 as questionable. In addition, three genomic islands were identified in the reference genome. The first, which extends from positions 3,433,545 to 3,495,693, is linked to the type IV secretion system and various virulence factors. Regarding key virulence loci, the presence of virulence-associated integrative conjugative element of *K. pneumoniae* variant 4—ICEKp4 and yersiniabactin cluster 10—ybt10 clusters were found within this island in the reference genome. Conversely, our results using Kleborate revealed the existence of ICEKp4, associated with the ybt10 cluster, at the identical genomic site as ICEKp3 in seven isolates characterized as bla_{KPC}-*negative* and belonging to ST45 (6/7). Furthermore, an unknown ybt was found in a bla_{KPC}-*positive* isolate (KPEC34K) with ST45-1LV. The second genomic island ranged between 3,555,090 and 3,607,401, encompassing genes encoding multidrug efflux pumps and metabolic enzymatic functions. The third genomic island, ranging from 4,502,849 to 4,545,387, predominantly harbors genes encoding hypothetical proteins, with some genes encoding phage integrase, transposase insertion sequence, DNA-binding prophage protein, and antirestriction protein (Supplementary Table 9). Remarkably, these two genomic islands were not present in the Ecuadorian isolates. In addition, comparison of Ecuadorian isolates with different reference genomes, showed variable distribution of gaps, some of these include prophage regions, suggesting a high genome plasticity of *K. pneumoniae* due to prophages (Supplementary Fig. 3).

Pangenomic analysis

To determine the genome architecture and diversity, a pangenome analysis was performed on the 14 K. *pneumoniae* genomes assembled in this study. The analysis revealed that 8,860 gene families were associated with the pangenome and were distributed among 3,511 core genes, zero soft-core genes, 2,332 shell genes, and 3,017 cloud genes. For each strain, there was an average of 5,097 genes with a slightly different number of identified protein-coding genes, with a minimum of 4,994 genes found in the KPEC9K isolate and a maximum of 5,677 genes found in the KPEC34K isolate. It is crucial to approach unique genes with caution, particularly when considering the low coverage obtained for the KPEC34 isolate, because our isolates presented an average of 186 unique genes. Pangenome analysis revealed that 177 new genes were added to each genome (Fig. 3A, 3C), and the outline curve did not reach a plateau, with a power-law exponent of 0.44 (Fig. 3B), indicating an open pangenome and increased genetic diversity among these clinical isolates.

Overview of antibiotic resistance genes

A clustering analysis was performed considering the presence of genes related to antibiotic resistance, *KPC*, and virulence factors. The presence of three nonrandom groups and ten subgroups can be observed (SIMPROF $p = 0.01$ at 999 iterations): the first Group A (Group A, 1 isolate) corresponded to the KPEC34K isolate with the presence of bla_{KPC-3}, while the second group registered a greater presence of genes related to virulence factors and absence of bla_{KPC} (Group B, 7 isolates—SIMPROF) and the third group (Group C, 6 isolates) with fewer genes related to virulence factors and the presence of bla_{KPC-2}, with the exception of KPEC9K. Importantly, the KPEC11K isolate was bla_{KPC-2} but did not show phenotypic resistance to carbapenems.

Categorization of the isolates was determined based on MLST by considering the presence or absence of

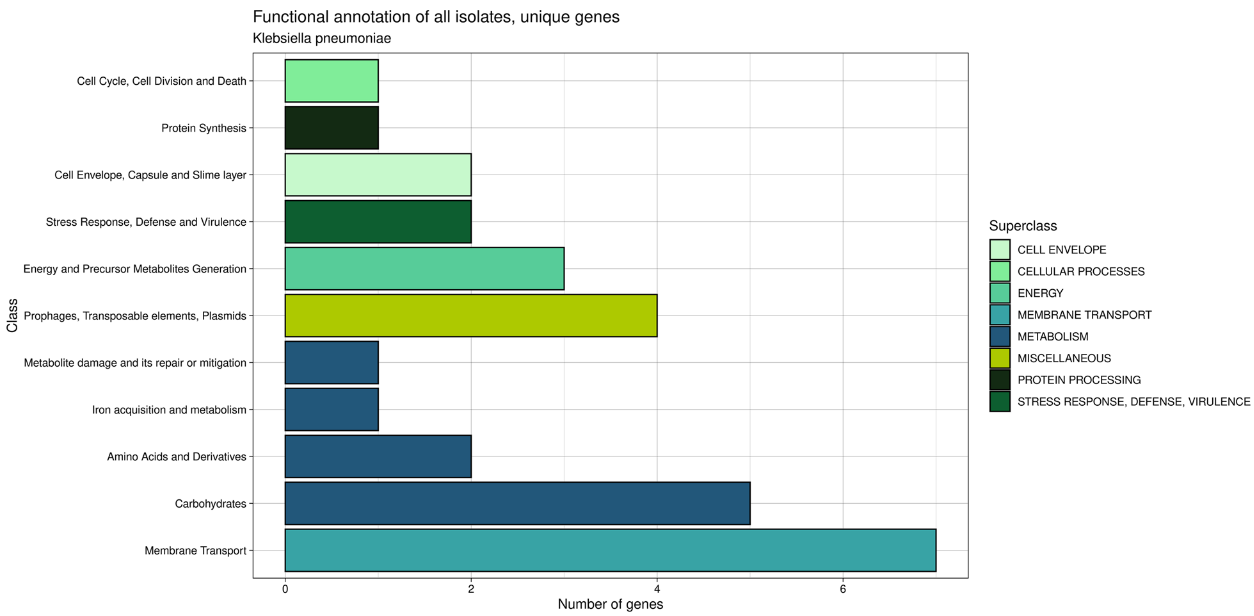

Fig. 1 Unique genes present in the functional annotation analysis of 14 K. *pneumoniae* Ecuadorian genomes

antibiotic resistance and virulence genes, revealing a predominance of ST45 in group B and ST629 in group C. A total of 44 resistance genes were identified in the isolates and classified into 10 antibiotic classes based on the Antibiotic Resistance Ontology (ARO) classification system. The antibiotic classes included β-lactams, aminoglycosides, fluoroquinolones, sulfonamides, trimethoprim, fosfomycin quinolone, phenicol, macrolides, and tetracycline (Fig. 4). The predominant resistance mechanisms observed included antibiotic inactivation (*aph(3”)-Ib*, *aph(6)-Id*, *aac(3’)-IIe*, $bla_{CTX\text{-}M\text{-}15}$) in a range–9–13 isolates, antibiotic target replacement (*sul2, dfrA14*) in 13 isolates, antibiotic target protection (*qnrB1*) in 10 isolates, and antibiotic efflux (tet(A)) in 9 isolates.

Analysis of the groups within the cluster revealed that group A exhibited 13 genes, whereas groups B and C had 18 and 27 genes, respectively. Unique genes were identified within each group, with nine unique genes in group A, four in group B, and 14 in group C. The principal resistance mechanisms exhibited by these genes primarily involved antibiotic inactivation, such as $bla_{KPC\text{-}3}$, $bla_{OXA\text{-}9}$, *aadA1, aac(6’)-Ib-AKT, aac(3)-IVa, aph(4)-Ia* in group A; $bla_{SHV\text{-}145}$, $bla_{TEM\text{-}1}$, $bla_{OXA\text{-}1}$, $bla_{SHV\text{-}101}$ in group B; and $bla_{SHV\text{-}11}$, $bla_{CTX\text{-}M\text{-}12}$, $bla_{KPC\text{-}2}$, $bla_{SHV\text{-}110}$, $bla_{SHV\text{-}106}$, fosA6, and mph(A). Additionally, antibiotic efflux has been identified as a key resistance mechanism in groups A and C (oqxAB alleles).

Our study examined resistance-associated point mutations (RAPM) in the isolates, identifying mutations primarily in *ompK35*, *ompK36*, *ompK37*, *gyrA*, *parC*, and *acrR* (Supplementary Table 10, Supplementary Fig. 1). For one hand, mutations in *ompK* genes affect β-lactam antibiotic resistance in non-carbapenemase-producing. Regarding *ompk35*, we found a premature stop codon in one of the isolates (KPEC15K), while in *ompk36*, eleven isolates had mutations mainly p.N49S and p.L59V (11/14) and 3 had the absence of the gene; finally, for *ompk37*, 13 isolates had mutations, predominantly p.I170M and p.I128M, while one had the absence of the gene. Notably, among seven carbapenem-resistant isolates in the antibiogram, three were blaKPC-negative but had porin mutations conferring resistance to this antibiotic class. On the other hand, our study showed that the *acrR* gene was present in 13 out of 14 isolates (92.85%), with one isolate (7.14%) lacking this gene. All isolates with the *acrR* gene shared the same mutations (*p.P161R, p.G164A, p.F172S, p.R173G, p.L195V, p.F197I, and p.K201M*), suggesting resistance to fluoroquinolones and β-lactams. Additionally, we detected a one-point mutation in *gyrA,* which was present in all the KPC-2 isolates (*p.S83I*). Similar to *gyrA*, *parC* was mutated (*parC* p.S80I) in KPC-2. Mutations in *parC* and *gyrA* are associated with resistance to fluoroquinolones, particularly ciprofloxacin.

We also identified the presence of cross-resistance genes, including the *oqxAB* system, composed of *oqxA* and *oqxB* subunits, and the *aac(6’)-Ib-DY18* (*aac(6’)-Ib-cr*) gene. These genes are known to elevate the minimum inhibitory concentrations (MIC) of specific fluoroquinolones. The *aac(6´)-Ib-cr* gene notably enhances aminoglycoside activity and induces cross-resistance to fluoroquinolones. We found both *oqxAB* and *aac(6’)-Ib-cr* genes in six isolates (KPEC14K, KPEC13K, KPEC15K,

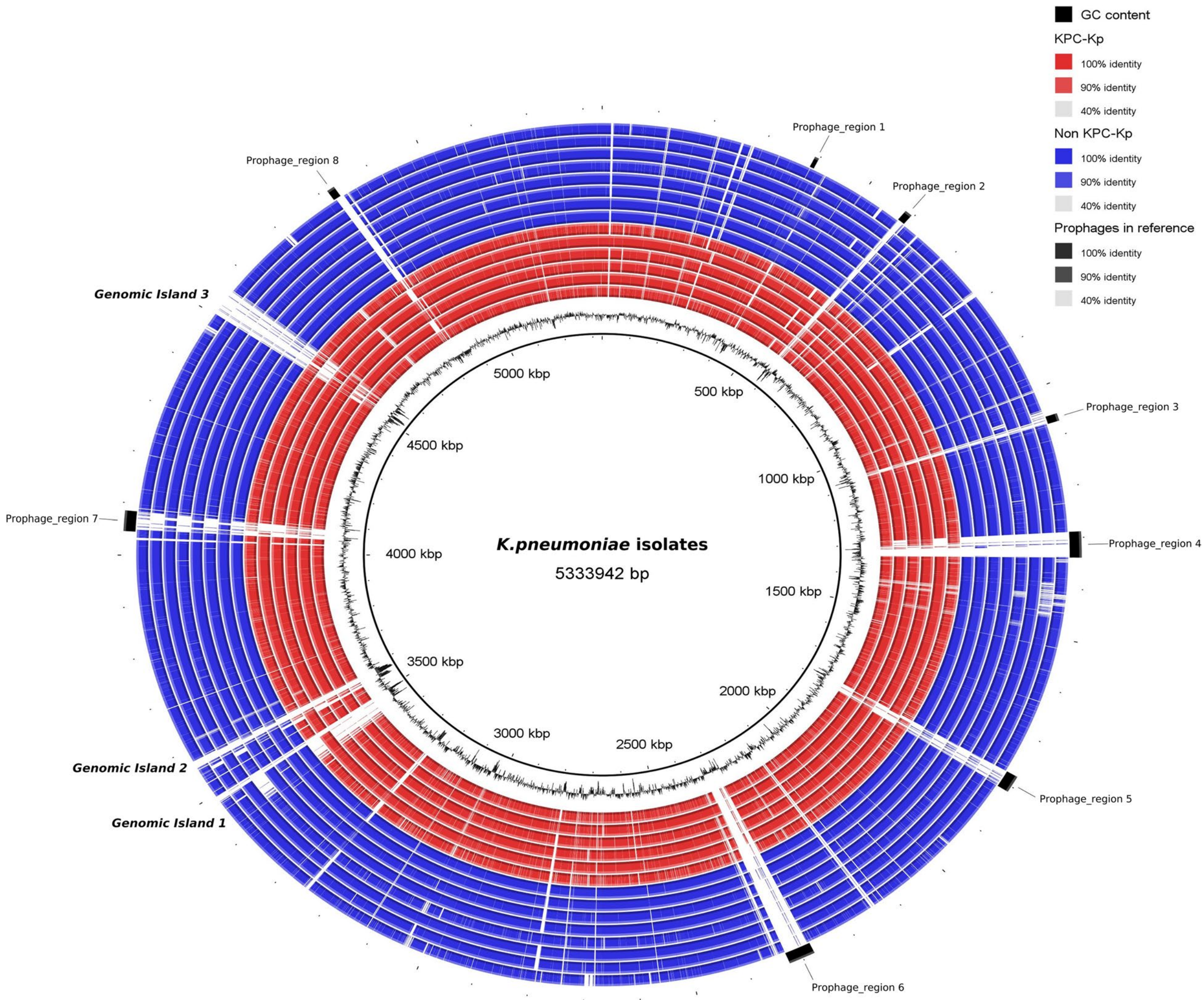

Fig. 2 Comparative genomic analysis of *K. pneumoniae* isolates using BLAST Ring Image Generator (BRIG). *The* bla_{KPC} (red circles) and non-bla_{KPC} (blue circles) isolates were compared to the reference *K. pneumoniae* HS11286 genome (inner black circle). The inner black section represents GC content. The inner circle represents the genome sequence of each sequenced strain. Black rectangles indicate prophages in the outer layer of the circles

KPEC28K, KPEC37K, and KPEC29K), suggesting a possible increase in fluoroquinolone resistance (Fig. 4).

Additionally, we examined the *acrAB* system, which imparts resistance to quinolones and beta-lactams, requiring both *acrA* and *acrB* subunits. Ten isolates had the complete system (Supplementary Fig. 1), while three isolates contained only one gene. One isolate was missing the acrAB system entirely.

Virulence factors

In this study, we analyzed 14 K. *pneumoniae isolates* using the Virulence Factor Database (VFDB) and identified a range of virulence factors, categorizing them into eight distinct groups based on their functional roles. We identified genes associated with biofilm formation (*mrkABCDFHIJLM*), adherence (*fimABCDEFGHIK*), immune evasion, secretion systems (*tssABCDFGHIJKLM*, *tli1*, and *KPHS_23120*), and a notably high number of genes associated with the iron-uptake system (*entABCDEF*, *fepABCDG*, *fes*, *iroE*, *ybtSXQPAUTE*, *irp1*, *irp2*, and *fyuA*). Additionally, a small number of genes were linked to efflux pumps (*acrAB*) and regulatory functions (*rcsAB*), with some isolates showing genes associated with serum resistance (Supplementary Table 11). The most prevalent virulence factors are related to the iron uptake system. Notably, the majority of the ST45 types possessed most of the iron uptake genes, whereas the yersiniabactin cluster

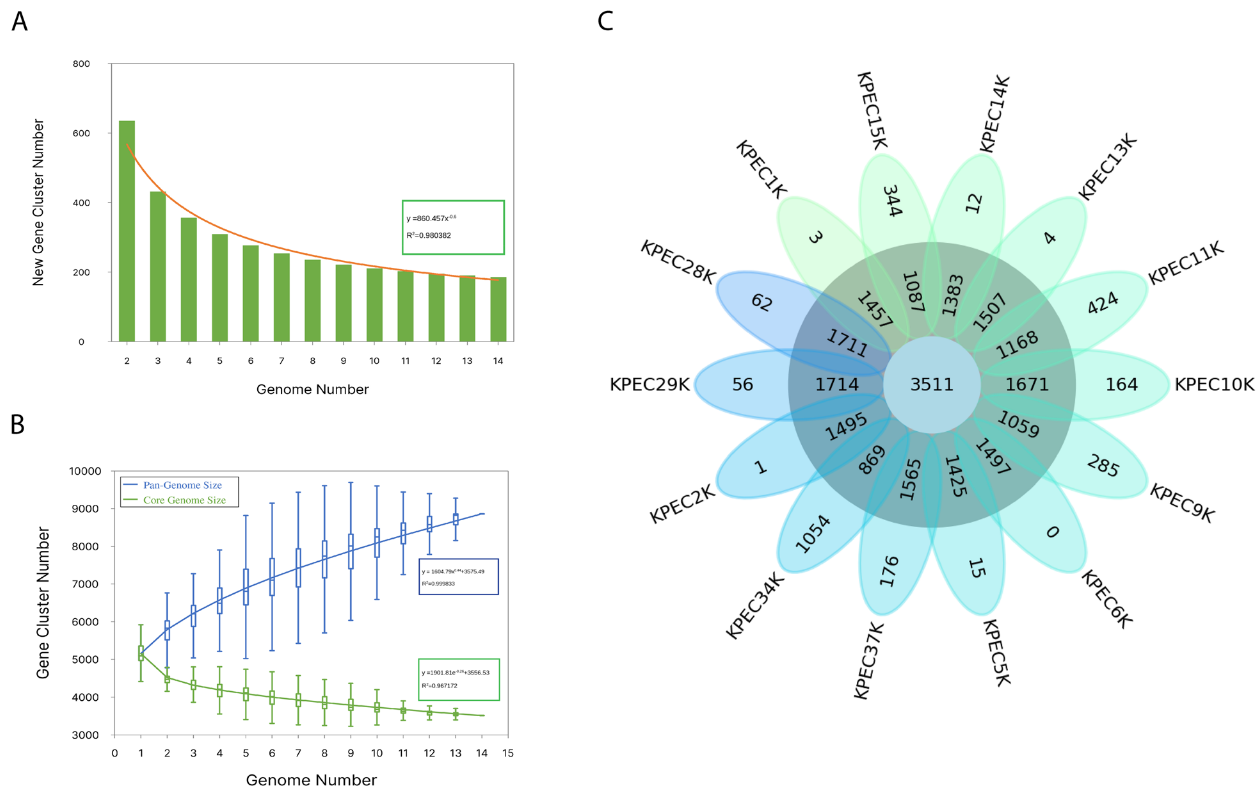

Fig. 3 Pangenome and functional annotation of *K. pneumoniae*. **A** Bar graphs showing the number of new genes associated with an increase in the number of *K. pneumoniae* genomes. **B** Gene accumulation curves of the pangenome (blue) and the core genome (green). Blue and green boxes denote the *K. pneumoniae* pangenome and core size for each genomic comparison, respectively. **C** Flower plots showing the core, accessory, and strain-specific genes of 14 K. *pneumoniae* isolates. The flower plot shows the core gene number (in the center), accessory gene number (in the annulus), and strain-specific gene number (in the petals)

(*ybtSXQPAUTE, irp1, irp2,* and *fyuA*) was found only in non-KPC isolates (Group B: SIMPROF). The second representative virulence factor is the secretion system, which includes many genes encoding the type VI secretion system (T6SS) that exhibits antibacterial activity (*tssA to tssL*).

In relation to Virulence-related biofilm genes, cps (capsular-encoded genes; see Fig. 4), *mrk* (type III fimbriae), and *fim* (adherence) genes, were identified in all the isolates. Regarding *non-KPC* strains, genes related to serum resistance and LPS synthesis (i.e. *wbbM* and *wzm*) were identified in this group. We also identified multidrug efflux pump transporters that are widespread among Gram-negative bacteria, including those of the resistance-nodulation-division (RND) superfamily, which is the most important mechanism associated with resistance against various antimicrobial molecules in *K. pneumoniae*. The *acrA* and *acrB* genes, which were identified in most of the Ecuadorian isolates (10/14), were classified into the RND family (Fig. 4). We also identified *rcsA* and *rcsB*, which are associated with capsule polysaccharide formation in most isolates. In terms of genes related to hypervirulence (e.g. aerobactin, salmochelin, etc.), none of them were not found in these isolates.

Plasmid prediction and resistance genes association

A total of 10 replicons were identified in our study, with ST45 isolates exhibiting ColRNAI, pKPN3, IncFII(pECLA), and IncFII_1_pKP91 replicons. In contrast, the IncL/M(pMU407) replicon was found only in the ST629 and ST629-LV isolates. Furthermore, the Col440I replicon was uniquely present in $bla_{\text{KPC-2}}$ isolates, whereas $bla_{\text{KPC-3}}$ isolates harbored distinct replicons, specifically FII (pBK30683) and IncFIA (HI1). Through plasmid reconstruction, our results revealed the association of specific resistance genes with certain isolates, such as the presence of a unique replicon of KPEC34K, FII(pBK30683) (accession number KF954760) (coverage 99.9%, pairwise identity 98.7%), which was associated with the presence of genes such

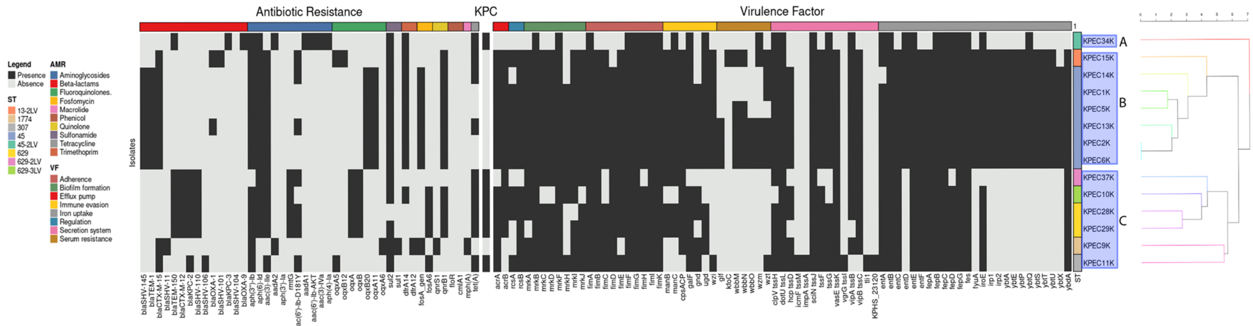

Fig. 4 Heatmap of antibiotic resistance genes and virulence factor profiles of 14 *K. pneumoniae* isolates from neonatal clinical isolates from Ecuador. Black boxes indicate the presence of genes, and gray boxes indicate the absence of genes. The characteristics of antibiotic resistance and virulence factors are represented in different colors in the first row, and these characteristics are described in the legend. Cluster analysis was based on the Euclidean distance between the genes present in each sample and representing real groups ($p=0.01$ at 999 iterations, SIMPROF)

as bla_{KPC-3}, *aac(6')-Ib-AKT, aadA1, dfrA14*, $bla_{\text{TEM-150}}$, *aph(6)-Id*, *aph(3'')-Ib*, *sul2*, and bla_{OXA-9}. Similarly, reconstruction of the pKPN-IT plasmid (accession number JN233704.1) revealed the resistance gene profile of the KPEC9K isolate, including *mph(A)*, *dfrA12*, and *aadA2*, which are known to be part of a class 1 integron (Fig. 5). Notably, although the pKPN3 plasmid was similar to pKPN-IT (coverage 76.2%, pairwise identity 99.3%), the latter contained an integron with resistance genes absent in pKPN3 isolates. None of the remaining replicons showed any association with the resistance genes in our isolates.

Phylogenetic analysis

Using the core single nucleotide polymorphisms (cgSNPs) of 90,474 SNPs, we conducted a phylogenomic analysis to determine the relationship between Ecuadorian isolates and isolates from other countries based on factors such as associated K-type, MLST, and the presence of carbapenemase-producing genes. According to geographical distribution, seven were related to North America and four were related to Africa, except for three isolates (KPEC15K, KPEC9K, and KPEC11K) that were distributed among isolates from Asia and Africa (Fig. 6). Among the MLSTs, ST45 and ST629 were the predominant STs among all Ecuadorian isolates considered in this study. However, the isolates from Ecuador clustered into different clades, including the ST45 and ST629 variants. Most ST45 isolates were non-KPC producers, in contrast to the ST629 isolates, which were KPC-positive. The remaining isolates showed a high level of diversity without discernible patterns. Additionally, the Ecuador isolates did not exhibit a link with those globally producing carbapenemases. Finally, we observed variations in the associated K-types across isolates.

Discussion

Whole-genome sequencing has provided a comprehensive view of the genomic content and epidemiological monitoring of clinical isolates of multidrug-resistant *K. pneumoniae*. Following the establishment of the AMR surveillance system in Ecuador in 2014, interesting discoveries were made. These findings involved the identification of high-risk strains, notably the New Delhi metallo-β-lactamase (NDM) strain of *K. pneumoniae,* in 2016 [52] and the isolation of an ST307 sequence variant carrying the bla_{OXA-48} like carbapenemase plasmid in 2020 [24]. Dedicated endeavors within the nation to enhance the monitoring of clinical *K. pneumoniae* strains are important. These efforts serve as a cornerstone for devising novel strategies for effectively managing local outbreaks. Using WGS, our study revealed key aspects related to the high resistance to commonly used antibiotics, virulence factors, and genomic characteristics of *K. pneumoniae* isolates from one of the most important pediatric hospitals in Guayaquil, Ecuador.

The identification of various sequence types (STs) and K-types revealed genetic heterogeneity within these isolates, with ST45 and K62 as the dominant types, harbored a high number of resistance genes and virulence factors in non-KPC isolates, indicating adaptation in the local hospital in neonatal intensive care units. Its presence has been previously reported in medical devices and infrastructure environments, revealing rapid dissemination of ST45 in these intrahospital environments [53]. Furthermore, it has been reported as one of the dominant carbapenemase-producing clones in the EuSCAPE surveillance study [32], whereas ST45 has been linked to bla_{KPC2} production in Venezuela [54]. This indicates that the variability in carbapenemase resistance profiles exhibited by sequence types stems from the acquisition of diverse plasmids containing resistance genes, genetic mutations affecting antibiotic susceptibility, and the impact of environmental selective pressures favoring certain resistant strains.

Sequence types associated principally with *bla*$_{KPC-2}$ isolates were identified (ST629 and its variants and ST307), and these findings are congruent with those of previous studies in Latin America [55, 56], in which ST629 was reported in others as one of the most prevalent in hospital patients [56], animal food, and the feces of wild birds [57]. This evidence supports the hypothesis that animal-adapted strains interact with humans [56]. ST307, which is considered a high-risk clone [58, 59], has been previously reported in a *bla*$_{OXA-48}$ positive isolate in Ecuador [24]. Notably, a low-frequency sequence type (ST1774) was identified in an isolate positive for *bla*$_{\text{CTX-M-15}}$ and *bla*$_{\text{SHV-11}}$, which only exists in a limited number of reports from turkey meat [60] and a clinical isolate from Germany [61]. These findings highlight the importance of genomic surveillance for tracking high-risk and emerging antimicrobial resistant clones, and the presence of various capsular types further emphasizes the complexity of the *K. pneumoniae* strains that could be circulating in the country.

Comparative genome analysis of the 14 K. *pneumoniae* isolates provided an overview of the genetic diversity within this species. The pangenome with an open structure in the Ecuadorian isolates indicated ongoing genetic heterogeneity among the clinical isolates and confirmed the plasticity that exists within the species [62]. Functional annotation revealed a greater abundance of genes related to metabolism, possibly indicating greater fitness during the progression of infection in the bloodstream, possibly through purine metabolism and vitamin biosynthesis [63]. These genes might enhance the pathogen capabilities of *K. pneumoniae* through different phases of virulence evolution, similar to what is observed in the metabolism of fucose [64]. In accordance with Shen et al. [65], our findings also revealed a higher prevalence of prophages in the reference genomes when compared to the Ecuadorian isolates. This suggests that the main distinguishing factor between the reference sequence and the aligned sequences is the presence of phages. Furthermore, these prophages are seen to play a significant role in the genome plasticity of *K. pneumoniae* [65].

Within the context of public health in Ecuador, the occurrence of *K. pneumoniae* infection in pediatric hospitals and the high prevalence of resistance underscore the escalating issue of MDR [66]. Most of the isolates in our study were resistant to commonly used antibiotics, including Ampicillin-sulbactam, ceftriaxone, and gentamicin, in addition the presence of resistance against cefepime, doripenem, cefotaxime, nitrofurantoin, and Trimethoprim–sulfamethoxazole emphasize the evolving antibiotic resistance mechanism. Although the number of samples analyzed in this study was limited, we found that Doripenem, Cefotacime, Nitrofurantoin, and Trimethoprim-sulfamethoxazole may be alternative options for the treatment of pediatric *K. pneumoniae* infections in local environments.

Previous studies using conventional molecular tools in Ecuador reported that *bla*$_{\text{KPC-5}}$ and *bla*$_{\text{KPC-2}}$ were the most common variants from 2010 to 2016 [17–19]. In the present study, 35.7% of the isolates presented the *bla*$_{\text{KPC-2}}$ gene, and one clinical isolate of *K. pneumoniae*

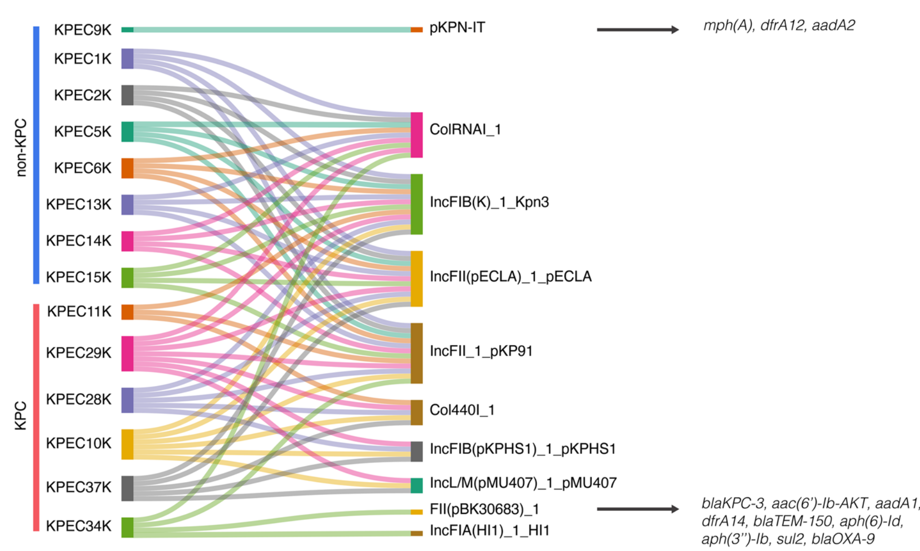

Fig. 5 The Sankey diagram shows the distribution of plasmid replicons between isolates and the far-right possible resistance genes associated with the presence of a replicon

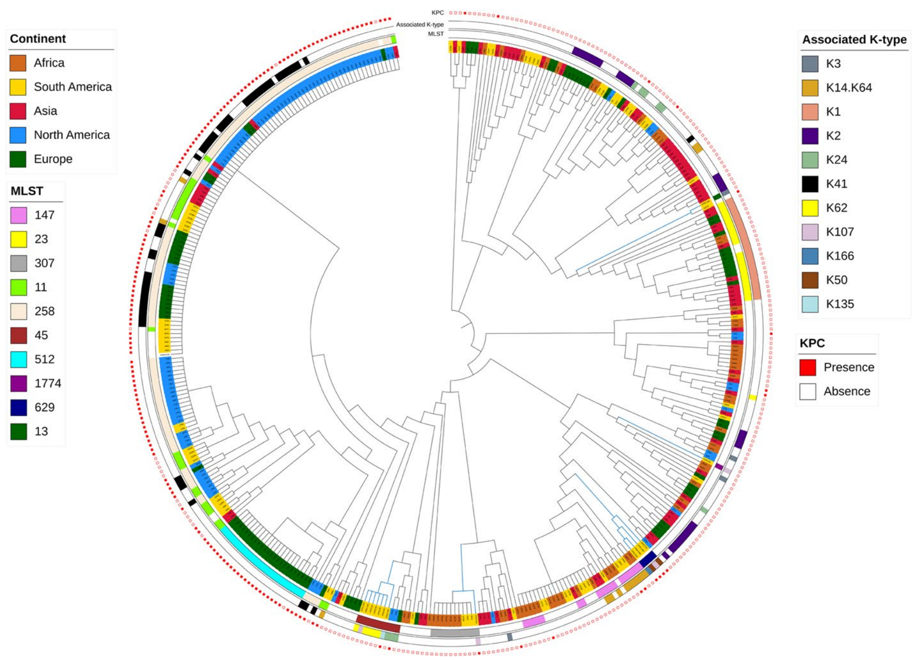

Fig. 6 Phylogenetic tree of *K. pneumoniae* isolates. Isolates from different phylogroups are colored differently. Layers around the tree denote (from inside out) MLST including sLV (ST locus variant), KPC, and the associated K-type. Blue branch color represents Ecuadorian isolates. Branch lengths are not to scale

was identified as *a* $bla_{KPC\text{-}3}$ gene producer and classified as ST45-2LV type, making the first clinical identification of *K. pneumoniae* $bla_{KPC\text{-}3}$ gene producers in Ecuador. $bla_{\text{KPC-2}}$ and $bla_{\text{KPC-3}}$ isolates have been found to be the main KPC producers across the USA, Europe, China, and Latin American countries, where KPC variants are endemic or cause outbreaks [67, 68], suggesting that KPC-3 can be disseminated in Ecuador.

Most carbapenemase-producing bacteria acquire resistance genes via plasmids [69], emphasizing the importance of advanced techniques to understand gene dissemination among bacterial species [70]. Despite challenges in the reconstruction of large plasmids using only short-read sequencing data, our study identified key AMR genes correlated with the presence of plasmid replicons, pKPN-IT plasmid in non-KPC isolates, and FII (pBK30683)_1 plasmid in $bla_{KPC\text{-}3}$ isolates. Therefore, to understand the correlation between AMR genes and plasmids, it is necessary to expand the number of isolates to include those circulating in Ecuador, and long-read sequencing data can be used to reconstruct whole plasmid sequences.

A comprehensive analysis of antibiotic resistance genes exhibited by *K. pneumoniae* has revealed an intricate spectrum of mechanisms contributing to the development of multidrug resistance. This study highlights the prevalence of critical resistance genes, including β-lactamase, aminoglycoside, and fluoroquinolone resistance-associated genes. Furthermore, the detection of cross-resistance genes, such as *oqxAB* and *aac (6')-Ib-cr* compounds, is complex and poses additional challenges in achieving effective antimicrobial treatment. Moreover, it is crucial to understand that not only the presence of resistance genes indicates possible resistance but also the presence of target-site point mutations confers resistance.

Several isolates identified as KPC producers based on phenotypic testing did not exhibit a genetic profile consistent with this phenotype, while some non-KPC isolates presented genes related to carbapenem resistance. These discrepancies highlight the complexity of the antibiotic resistance mechanisms in these bacteria and their impact on treatment, as antibiotic therapy is selected through susceptibility testing. In our study, $bla_{\text{CTX-M-15}}$*was* present in 64% (9/14) of the isolates and co-occurred with mutations in the *ompK36* protein, potentially contributing to carbapenem resistance [71, 72]. In addition, we observed a correlation between specific point mutations and STs. For instance, the *ompK37* p.N230G mutation appears to be related to ST45 and ST307 [73], a pattern that is also evident in Ecuadorian isolates with the same ST and its locus variant. This combination likely explains the resistance mechanisms of isolates that displayed a KPC-positive phenotype but lacked a matching genotype. Mutations or deletions in ompK genes, which encode porins, and the presence of typical carbapenem resistance genes can significantly impact the resistance profile of *K. pneumoniae*.

Such insights underscore the need for a more comprehensive approach in the genetic analysis of antibiotic resistance, considering not only the presence of resistance genes, but also the potential impact of other genetic alterations, such as resistance-associated point mutations.

Concerning virulence factors, we identified 76 genes related to adhesion, biofilm formation, immune evasion, iron uptake and others, grouped into 9 loci. These factors underscore the potential of these strains to establish infections and evade host defense. Most of our isolates exhibited type 1 and type 3 fimbriae. However, we also identified a distribution of virulence genes that appeared to be associated with ST. Focusing on biofilm formation and adhesion virulence factors, we identified the *mrk* gene cluster, which encodes type 3 fimbriae [74]. The key structural components, *mrkA* and *mrkD*, were absent in the KPEC34K isolate, indicating a loss of type 3 fimbrial functionality. For iron uptake, ST45 isolates harbored genes involved in the biosynthesis of yersiniabactin, a key virulence factor. This was consistent with the findings of other studies, in which yersiniabactin was detected in more than 83% of ST45 isolates [75]. Additionally, we found that most of the non-KPC isolates carried genes that encode more than one siderophore, enterobactin, or yersiniabactin, which made these isolates more virulent, as reported in other studies [76]. We identified virulence factors related to immune modulation, such as the capsule and LPS. These genes are related to inactivation of inflammation, inhibition of complement activation, prevention of phagocytosis, opsonophagocytosis, and other processes [74]. We also observed a correlation between the O-locus, one of the major components of LPS, and MLST, revealing that ST45 was related to the O2a locus, ST629 was related to O3/O3a, ST1774 was related to O12, ST307, and ST13-2LV was related to O2afg. A similar situation occurred in other studies, where O2a was more related to ST45, and O2afg was more related to ST307 [77]. This study highlights the complexity and diversity of virulence factors in *K. pneumoniae*, revealing the importance of these genes in pathogenicity and the adaptability of these strains for establishing infections in hospital environments.

Our study has several limitations. First, we could not determine the epidemiological patterns in the local area because of the limited number of isolates and the low number of representative isolates per site and year sequenced for this study. Second, because of the limitation in the type of sequencing, it was not possible to validate the presence of plasmids associated with antimicrobial resistance; therefore, a combination of short- and long-read sequencing technologies was needed. Third, there was incomplete phenotypic information on antimicrobial resistance in the antibiogram.

Conclusions

The present study showed the wide genomic diversity of *K. pneumoniae* isolates from a pediatric hospital in Guayaquil-Ecuador, characterized by WGS. We found substantial intraspecies diversity in the isolates, as evidenced by MLST and capsular typing, with ST45 and K62 being prevalent in non-KPC isolates, and ST629 in KPC. Analysis of the pangenome revealed high genetic diversity among Ecuadorian clinical isolates. Furthermore, functional classification has identified several structural complexes, highlighting the metabolic and virulence potential of this pathogen. These findings underscore the importance of understanding the diversity of *K. pneumoniae* isolates to develop effective strategies to prevent and treat infections caused by this pathogen. Additionally, in-depth epidemiological studies could provide valuable insights into the transmission dynamics and factors contributing to the dissemination of multidrug-resistant *K. pneumoniae* strains in our country.

Abbreviations

SAM	Ampicillin-sulbactam
CRO	Ceftriaxone
GEN	Gentamicin
FEP	Cefepime
CIP	Ciprofloxacin
DOR	Doripenem
IPM	Imipenem
MEM	Meropenem
TZP	Piperacillin–tazobactam
AMK	Amikacin
FOF	Fosfomycin
CTX	Cefotaxime
NIT	Nitrofurantoin
SXT	Trimethoprim–sulfamethoxazole
* SDD	Susceptible-dose-dependent
MIC	Minimum inhibitory concentration

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12864-024-10835-9.

Additional file 1: Supplementary Figure 1. Phylogenomic analysis of Ecuadorian isolates against a reference revealed MLST, K type and the presence/absence of several important genes related to point mutations and cross-resistance.

Additional file 2: Supplementary Figure 2. For the functional annotation of K. pneumoniae from 14 KPEC isolates, the x-axis represents the average number of genes per isolate, the y-axis represents the class assignment, and the color represents the superclass.

Additional file 3: Supplementary Figure 3. Comparative genome analysis of *K. pneumoniae* of Ecuadorian isolates with the reference genomes. Prophages are indicated in black bars on the outside circle. The reference genome is located on the innermost side.

Additional file 4: Supplementary Table 1. Characteristics of *K. pneumoniae* clinical isolates from Ecuador.

Additional file 5: Supplementary Table 2. Depth sequencing of antimicrobial resistance genes.

Additional file 6: Supplementary Table 3. Genomes from Pasteur Institute using in phylogenomic analysis.

Additional file 7: Supplementary Table 4. Genomic characteristics of *K. pneumoniae* clinical isolates.

Additional file 8: Supplementary Table 5. O-locus information of 14 isolates of *K. pneumoniae*.

Additional file 9: Supplementary Table 6. Number of functional genes per isolate and superclass by PATRIC annotation.

Additional file 10: Supplementary Table 7. Functional annotation by PATRIC of all 14 *K. pneumoniae* genomes.

Additional file 11: Supplementary Table 8. Prophage coverage per isolate.

Additional file 12: Supplementary Table 9. Genomic island identified in the *K.pneumonie* reference genome HS11286.

Additional file 13: Supplementary Table 10. Analysis of antimicrobial resistance determinants ordered by *K. pneumoniae* isolates.

Additional file 14: Supplementary Table 11. Presence of virulence factor genes in Ecuadorian isolates.

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Authors' contributions
I.M.L and D.A.M conceived and planned the study. J.C.H.O, J.G.C.A and M.R performed sampling and DNA extractions. G.M.L, J.C.F.C and D.A.M performed sample processing (library prep and sequencing). I.M.L, G.M.L and J.C.F.C performed genome assembly, annotation, and comparative genomic analyses. I.M.L, D.A.M wrote the draft manuscript. D.A.M supervised the study and obtained funding. All authors reviewed the manuscript.

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Availability of data and materials
The Klebsiella pneumoniae whole-genome sequencing data will be deposited in the public archive of NCBI GenBank BioProject under accession number PRJNA956653, SRA data: SRX19999240, SRX21482265 to SRX21482277. For reviewers, we shared the sequence data: https://drive.google.com/drive/folders/10CRhiwmgYq4fykzJ98dTuJ8uPySSDH5d?usp=sharing.

Declarations

Ethics approval and consent to participate
This study was approved by the Ethics Committee of Universidad Espíritu Santo (CEISH-UEES) project N0 2022-001A. Anonymity and the protection of personal data were preserved. The clinical isolates described come from a hospital biobank and each isolate is unidentified, ensuring that it cannot be traced back to individual patients. The need of informed consent was waived by the CEISH-UEES Ethics committee.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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